

Auteur: Rubin Cuypers
Studierichting: Informatica
Oefeningenreeks 5D nr. 5.1

$$y^2 = 2x - 1 \text{ en } y = 2x - 3$$

$$x_1 = \frac{y^2 + 1}{2} \text{ en } x_2 = \frac{y + 3}{2}$$

$$\Rightarrow S = \int_{-1}^2 \left(\frac{y + 3}{2} - \frac{y^2 + 1}{2} \right) dy$$

$$= \int_{-1}^2 \frac{1}{2} (-y^2 + y + 2) dy$$

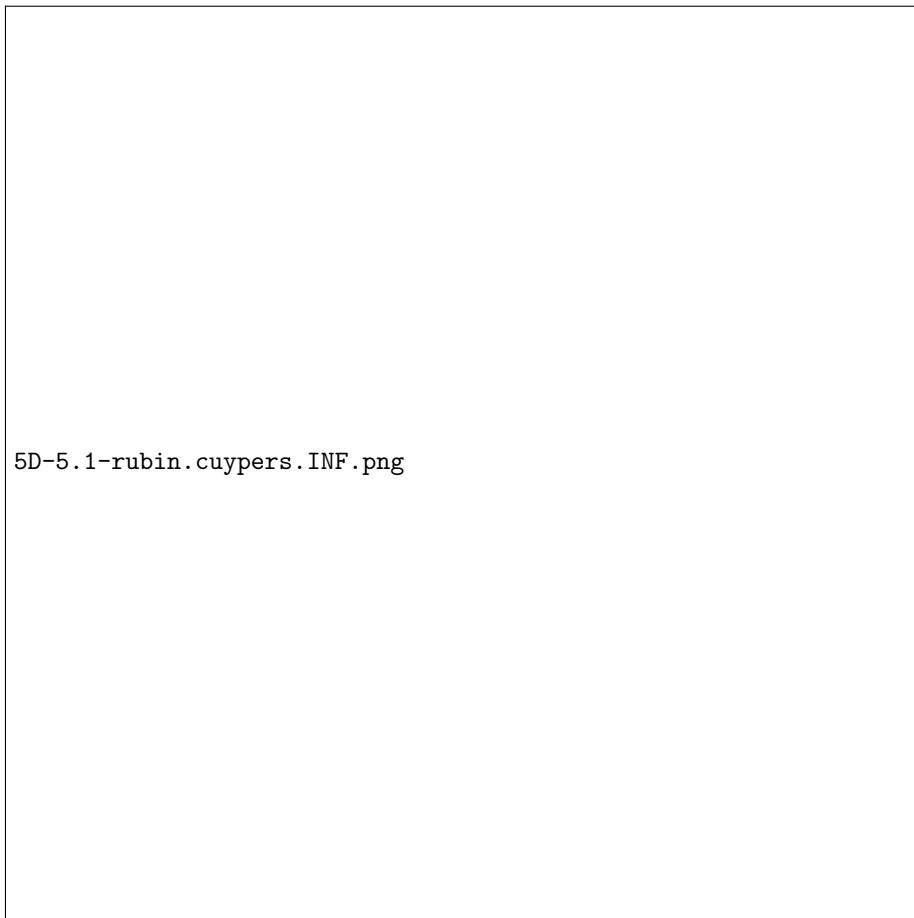
$$= \left[-\frac{y^3}{6} + \frac{y^2}{4} + y \right]_{-1}^2$$

$$= -\frac{4}{3} + 1 + 2 - \left(\frac{1}{6} + \frac{1}{4} - 1 \right)$$

$$= \frac{27}{12}$$

$$= \frac{9}{4}$$

References



5D-5.1-rubin.cuyppers.INF.png